

Ellie Madsen

EDUCATION

Columbia University Graduate School of Architecture, and Planning, and Preservation M.S. Computational Design Practices	2025-Present
Brown University B.A. Computer Science, B.A. Environmental Studies	2018-2022
New London High School, Wisconsin Valedictorian	2014-2018

RESEARCH

Research Assistant Center for Climate Systems Research Columbia University Climate School & NASA Goddard Institute for Space Studies	2026-Present
Research Assistant Visualizing Icesheets in Extended Reality (VISER) Columbia University Computer Graphics & User Interfaces Laboratory & Polar Geophysics Group	2025

TEACHING

Teaching Assistant Columbia University Graduate School of Architecture, and Planning, and Preservation <i>Physical Computation</i>	2026-Present
Teaching Assistant Brown University Department of Computer Science <i>Data Structures and Algorithms, Object-Oriented Programming, Socially Responsible Computing</i>	2021-2022

PROFESSIONAL

Software Engineer II Pinterest <i>Retrieval infrastructure, vector databases, distributed systems, cluster management, embedding quantization</i>	2021-2025
Intern City of Providence Office of Sustainability	2020
Library Assistant Brown University Sciences Library	2018-2022
Intern New London Engineering	2018-2019

AWARDS

Merit Scholarship Columbia University GSAPP	2025
Dean's Scholarship Columbia University GSAPP	2025
SPRINT Fellowship Institute at Brown for Environment and Society	2020
National Merit Scholar College Board	2018
Valedictorian New London High School	2018

TECHNICAL SKILLS

Coding Languages: C++, Java, Python, JavaScript, HTML, CSS
2D: Figma, Adobe Illustrator, Photoshop, InDesign
3D: Rhino3D, Grasshopper, Unity (VR, XR, MRTK), Three.js, PyVista, Blender, Solidworks
Geospatial: QGIS, ArcGIS, Google Earth Engine, GeoPandas, OpenStreetMap, Mapbox
DevOps: Git, AWS, JIRA, Kubernetes
Skills: geospatial analysis, computer graphics, data visualization, parametric modeling, machine learning, generative design, cloud computing, web development, UI/UX, VR, search infrastructure, system architecture, distributed systems, cluster management, vector databases, quantization, graph embeddings
Physical: Arduino, printmaking, sewing, lasercutting, woodworking, sculpture

YOGA

Yoga Instructor Dodge Fitness Center	2025-Present
200 Hour Yoga Teacher Training One Yoga, Thailand	2025